

Multiple Choice

1. **Answer:** D

Options: A) Neptune; B) Uranus; C) Saturn; D) all of the above

Note: Correct option: D - all of the above

2. **Answer:** A

Options: A) The Moon and Earth have nearly identical isotopic compositions.; B) The Moon's orbit is not in the same plane as Earth's equator.; C) The Moon's crust is thicker on the far side than the near side.; D) The Moon has a much smaller iron core than the Earth even considering its size.; E) The Moon was once entirely molten.

Note: Correct option: A - The Moon and Earth have nearly identical isotopic compositions.

3. **Answer:** B

Options: A) 750.0; B) 924.0; C) 1300.0; D) 800.0; E) 870.0

Note: Correct option: B - 924.0

4. **Answer:** A

Options: A) frequency does not change, the wavelength increases, and the speed increases; B) frequency decreases, the wavelength increases, and the speed does not change; C) frequency increases, the wavelength increases, and the speed decreases; D) frequency increases, the wavelength does not change, and the speed increases; E) frequency increases, the wavelength decreases, and the speed does not change

Note: Correct option: A - frequency does not change, the wavelength increases, and the speed increases

5. **Answer:** D

Options: A) 1000 Mev; B) 650 MeV; C) 800 MeV; D) 931 Mev; E) 500 Mev

Note: Correct option: D - 931 Mev

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1.9 eV'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '32.185 ftullets⁻²'.

7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '0.750 gm'.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'independent of M'.

Long Form Response

11. **Answer:** The U.S.S.R.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 600 J. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key